

1-Month Robotics Club Curriculum (SHS)

Format (2 hours each week): 10 min recap, 20 min demo, 75 min build challenge, 15 min debugging/code review.

Week 1: Arduino Refresh & GPIO

Quick refresher (15 min): Arduino architecture, pins, Serial Monitor.
Hands-on: Blink patterns, push buttons, RGB LED, buzzer challenges.
Challenge: Simon Says or reaction timer.

Week 2: Sensors

Quick refresher: digital vs analog inputs.
Hands-on: Ultrasonic, LDR, potentiometer, servo.
Challenge: Automatic night light or parking sensor.

Week 3: Actuators & Mini Systems

Hands-on: DC motors, motor driver, servo control, combining sensors with outputs.
Challenge: Obstacle alert system or simple motorized mechanism.

Week 4: Integration Challenge

Teams build a small system using at least one sensor and one actuator. Examples: Smart parking sensor, automatic door, security alarm.
Presentation and code review.

Club Expectations: Use Git or versioned sketches where possible, keep wiring neat, comment code, test modules individually, and ensure every team member can explain the solution.